

Superheroes Social Skills: A Comparison of Video Only and Full Curriculum on Social Skill Use

Focus on Autism and Other
Developmental Disabilities
2021, Vol. 36(2) 95–107
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DOI: 10.1177/1088357621989260
focus.sagepub.com



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Abstract

A substantial number of multicomponent interventions have been developed to address social functioning in individuals with autism spectrum disorder (ASD). For multicomponent interventions, identifying critical elements of the intervention related to behavior change is critical and may potentially improve the social validity of interventions. Such an analysis of the Superheroes Social Skills program was conducted. Three participants with ASD and one participant without a diagnosis participated in a 4-week intervention in which three target social skills were taught. Participants were first exposed to only the video-based elements of intervention, followed by participation in the full curriculum. Results of the study indicated that exposure to video elements often resulted in limited increases in skill accuracy, but the largest increases were observed during training with the full curriculum. Limitations and future directions are discussed with respect to multicomponent interventions for individuals with ASD.

Keywords

social skills, autism spectrum disorder, video modeling, superheroes social skills

Social skills instruction represents one of the most frequently implemented interventions for individuals with autism spectrum disorder (ASD; Hess et al., 2008). The frequency with which social skills interventions are implemented is likely, in part, due to parents and caregivers of children with ASD reporting that development of social skills represents an extremely high priority (Pituch et al., 2011; Sagers et al., 2019). Given the nature of social skills deficits to ASD, as well as the demand for intervention to promote social skills from parents and caregivers, a substantial number of interventions have been developed to promote social skill acquisition and use. Interventions evaluated for promoting social skills may consist of implementation of a single intervention strategy in isolation (e.g., video modeling; Lee, 2015)—with many strategies being found to promote acquisition and use of target social skills when used in this way (e.g., Dueñas et al., 2019). Alternatively, several empirically supported social skills intervention strategies may be implemented simultaneously, as commonly observed in manualized social skills interventions (e.g., Jenson et al., 2011; Laugeson et al., 2015; Sofronoff et al., 2015), as a means of enhancing intervention effects or promoting generalization of skills from one context to another.

The Superheroes Social Skills program (Jenson et al., 2011) is one example of a manualized intervention that

incorporates multiple empirically supported intervention strategies to address challenges related to generalization of social skills (e.g., Watkins et al., 2019). Specifically, Superheroes Social Skills includes behavioral skills training (Stewart et al., 2007), video modeling (Bellini & Akullian, 2007), and social narratives (Karl & Wolfe, 2018) as strategies for training target social skills. It is important to note that video models within Superheroes Social Skills differ slightly from traditional video modeling. Generally, traditional video modeling consists of an individual watching a video and then emulating the behavior depicted in the video (Bellini & Akullian, 2007; Nikopoulous & Keenan, 2007). However, video modeling within Superheroes Social Skills is more intricate, in that there are three separate video models for each corresponding target skill within the curriculum. The first video model

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consists of an introduction to the skill along with a rationale for using the skill. The second video model consists of unknown peers performing the target skill in various environments (e.g., at school, at home). The third and last video model consists of a digital comic, which is a type of animated social story that has superhero characters utilizing the target social skill in an original scenario. At the end of the digital comic, participants are presented with an opportunity to describe the steps that comprise the target skill.

The Superheroes Social Skills program targets 17 various discrete social skills ranging from Foundational (e.g., Following Directions) to Intermediate (e.g., Responding to Questions/Requests) to Advanced (e.g., Recognizing Emotions). Within the curriculum, each discrete skill has a corresponding task analysis that explains how to demonstrate an appropriate social skill. During each lesson, the group facilitator begins the session by welcoming participants to the group, followed by an introduction of the target skill with provision of rationale via video model, in vivo modeling of the skill is demonstrated by a group facilitator, followed by viewing the video model with unknown peers. Students then engage in role play of the skill and receive either corrective feedback or praise based on demonstration. Students finish the lesson by playing a social game that incorporates the use of the target skill and a reward is provided for participation during the session.

Previous research suggests positive outcomes for individuals with ASD who participate in the intervention, particularly in regard to improving demonstration of discrete social skills. For example, Murphy et al. (2018) demonstrated improvements in participants' accuracy of discrete skills, specifically Body Basics, Participate, and Expressing Wants and Needs, following the implementation of Superheroes Social Skills in both training and generalization settings. In addition, Radley, O'Handley, et al. (2014) demonstrated the effects of Superheroes Social Skills on improving participants' demonstration of target skills that included Participation, Conversation, Perspective Taking, and Problem Solving. Other studies have documented improvements in increased social engagement of participants with ASD during recess periods (Radley, Ford, et al., 2014), specifically. Outcomes have also suggested improved ratings of social functioning (Block et al., 2015) and increased connection with peers following participation in the program (Radley et al., 2017).

Despite research supporting the utility of the intervention in promoting social skills acquisition and use, the multielement nature of the intervention has limited previous research on the Superheroes Social Skills program. Specifically, the inclusion of multiple intervention components has limited the ability of researchers to identify which of the components is related to behavior change of participants. As such, it is unknown whether the multicomponent

nature of the Superheroes Social Skills provides additive benefit, or whether a single strategy within the packaged intervention (e.g., video modeling) would be sufficient in promoting behavior change while also representing a less intensive intervention approach.

The question of relative contribution of intervention elements, as well as the sufficiency of strategies to promote behavior change in isolation, is of utmost importance to practitioners. Specifically, intervention elements that do not contribute to improved outcomes of participants ultimately result in unnecessary expenditure of time and effort of those who elect to utilize the intervention. Relatedly, different components of a multielement intervention may require varying levels of response effort (i.e., some components requiring more time or preparation). When practitioners encounter such elements, they may elect not to utilize those elements and only partially implement an intervention. Similarly, when working with time constraints (e.g., a child is only available for a limited amount of time each week), practitioners may selectively implement intervention components as time allows. To provide an example, school-based users of the Superheroes Social Skills program could elect to only show video-based elements due to the limited amount of time that may be available to facilitate a weekly social skills group. Such a practice may result in outcomes that differ dramatically from those observed in evaluations of a multicomponent intervention in its entirety. Therefore, the rationale of this study was to evaluate the implementation of solely the video model component of the program to determine if this was sufficient to improve the accuracy of participants' social skills.

Although previous evaluations of the Superheroes Social Skills program have documented improvements in areas such as social skills accuracy, social engagement, and stakeholder ratings of social functioning, the multicomponent nature of the intervention limits the ability to determine which elements of the intervention are active in promoting behavior change. As such, it is unknown whether all elements of the intervention are necessary, or whether a modified version of the intervention in which some components are removed would be similarly efficacious. Ultimately, identifying active elements of the intervention is likely to result in increased social validity and efficacy of the intervention (Ward-Horner & Sturmey, 2012; Wolf, 1978), as more resources may be devoted to those elements of an intervention that are most associated with behavior change.

Given that researchers have yet to evaluate the relative contribution of individual intervention components within the Superheroes Social Skills program, this study sought to evaluate the relative contribution of those elements presented via video in relation to the full curriculum. In doing so, this study intended to determine whether a relatively less effortful version of the intervention consisting only

of video-based elements of the intervention would be sufficient in promoting an increase in accuracy of discrete social skills targeted within the program. Thus, this study sought to answer whether implementation of only the video component within the Superheroes Social Skills program results in increased target skill accuracy of elementary-age children with ASD.

Although this is not a true component analysis in that two or more isolated independent variables are not systematically analyzed (Ward-Horner & Sturmey, 2012), to the authors' knowledge, this will be the first evaluation in which an isolated component of Superheroes Social Skills is evaluated. This study aims to add to the literature by addressing the question regarding the necessity of implementing the entirety of Superheroes packaged intervention to produce mastery skill levels. This is important when considering school personnel's time, as a full lesson of Superheroes takes approximately 30 min to an hour to implement (Murphy et al., 2018; O'Handley et al., 2016) whereas implementing only the video component would be substantially shorter in duration. The following research question guided the present investigation:

Research Question (RQ1): Is a video-only version of Superheroes Social Skills as effective in promoting social skill accuracy as implementation of the full curriculum?

Method

Participants and Setting

Prior to recruitment of participants, all procedures were approved by the affiliate university's Institutional Review Board. Inclusion criteria for the study consisted of parent-reported social skills deficits and a chronological age between 5 and 9. Exclusion criteria included concurrent participation in other social skills intervention.

Participants were recruited via flyers that were posted at the university, local school districts, and local clinics. All participants were enrolled in the 1-month program by their parent or guardian, who provided consent to participate. Prior to the program start date, assent was also obtained from participants. All consenting participants were administered the Expressive Vocabulary Test, Second Edition (EVT-2; Williams, 2007); Peabody Picture Vocabulary Test, Fourth Edition (PPVT-4; Dunn & Dunn, 2007); and the Stanford-Binet, Fifth Edition (SB-5; Roid, 2003) Abbreviated Battery IQ (ABIQ) to determine linguistic and cognitive functioning. Parents completed the Autism Social Skills Profile (ASSP; Bellini & Hopf, 2007) which was utilized for skill identification and Autism Spectrum Rating Scale (ASRS; Goldstein & Naglieri,

Table 1. Participant Demographics.

Participant	Gender	Age	Race	Diagnosis
Otto	Male	8	White	ASD
Lars	Male	9	White	ASD
Trish	Female	8	White	ASD
Tito	Male	5	White	None

Note. ASD = autism spectrum disorder.

2009) to quantify symptoms and behaviors associated with ASD.

Participant demographic data and assessment data are presented in Tables 1 and 2. Otto was an 8-year-old White male with a diagnosis of ASD and a special education classification of Autism. Assessment data indicated below average cognitive functioning and language abilities, and the ASRS Total Score was slightly elevated. The ASSP indicated social skill functioning that was above the mean of the normative group (i.e., 105.82; Bellini & Hopf, 2007). Lars was a 9-year-old White male with a diagnosis of ASD and a special education classification of Autism. Assessment data indicated below average cognitive functioning and language abilities. Completion of the ASRS indicated an Elevated Total Score, and the ASSP indicated social functioning above the mean of the normative group. Trish was an 8-year-old White female with a diagnosis of ASD and a special education classification of Autism. Assessment data indicated cognitive functioning and language abilities within normal limits. The ASRS Total Score was in the elevated range, and the ASSP total was above the mean of the normative group. Trish was absent during intervention for the Emotions skill due to vacation. In considering assessment data, it is notable that all participants demonstrated ASRS scores above the normative group from Bellini and Hopf (2007), suggesting relatively fewer social skill deficits or detrimental social behaviors.

Tito was a typically developing peer who was enrolled due to parent-reported social difficulties that were consistent with participants with ASD and skills addressed within the Superheroes Social Skills program, and as a means of increasing exposure to social interaction prior to beginning kindergarten. Initially recruited as a peer, Tito did not undergo any preintervention assessment. However, Tito was subsequently included in data collection following researcher observation of levels of social skill deficits similar to that of peers with ASD. Although Tito's caregiver consented to data collection, they did not consent to the standardized assessments completed with other participants.

The summer social skills program took place in a university classroom containing two long tables and open floor space for children to play games during free time. There was also a projector and screen in the room which were

Table 2. Assessment Results.

Participant	SB-5 Abbreviated IQ (Description)	PPVT-4 Standard Score (Description)	EVT-2 Standard Score (Description)	ASRS Total Score (Description)	ASSP (Description)
Otto	73 (borderline impaired or delayed)	74 (moderately low)	74 (moderately low)	86 (slightly elevated)	129 (N/A)
Lars	52 (moderately impaired or delayed)	75 (moderately low)	76 (moderately low)	93 (elevated)	123 (N/A)
Trish	112 (high average)	110 (average)	99 (average)	96 (elevated)	124 (N/A)

Note. SB-5 = Stanford-Binet, 5th Edition; PPVT-4 = Peabody Picture Vocabulary Test, 4th Edition; EVT-2 = Expressive Vocabulary Test, 2nd Edition; ASRS = Autism Spectrum Rating Scale; ASSP = Autism Social Skills Profile. SB-5 descriptive ranges: 40–54 = moderately impaired or delayed, 55–69 = mildly impaired or delayed, 70–79 = borderline impaired or delayed, 80–89 = low average, 90–109 = average, 110–119 = high average, 120–129 = superior, 120–144 = gifted, 145–160 = very gifted; PPVT-4 and EVT-2 descriptive ranges: <70 = extremely low, 70–85 = moderately low, 86–114 = average, 115–130 = moderately high, >130 = extremely high; ASRS descriptive ranges: <40 = low, 40–59 = average, 60–64 = slightly elevated, 65–69 = elevated, 70+ = very elevated.

used to display videos. Sessions occurred two times per week and lasted approximately 2 hr. Intervention was conducted across 4 weeks, with all intervention sessions facilitated by doctoral-level graduate researchers.

Materials

The Superheroes Social Skills program was utilized during all intervention sessions. The program is a packaged intervention containing video models, comic book social narratives, and self-monitoring cards that are directly tied to each social skill. Furthermore, the intervention manual provides explicit instructions for implementation of the full intervention, inclusive of descriptions and rationales for use of target skills, as well as role-play scenarios for each skill. Additional play materials were utilized during free time and included items such as board games, LEGOs, coloring books, and building blocks.

Dependent Measure

The primary dependent variable was the percentage of skill accuracy for three target skills: Conversation, Emotions, and Responding to Questions. Skill accuracy was defined as the number of skill steps an individual demonstrated correctly divided by the total number of possible steps, multiplied by 100. Each target skill had a task analysis adapted from the Superheroes Social Skills manual, each of which was dichotomously marked as “demonstrated” or “not demonstrated” for each step following the presentation of a cue for skill use (see Table 3 for task analyses). Data were collected by trained graduate researchers who provided participants with opportunities to engage in the desired social skill. A researcher provided a cue that consisted of a statement (e.g., “How are you today?” “What do you think about sports?”) indicating that the participant was supposed to use a specific social skill. Following the presentation of the cue, a graduate researcher recorded whether the discrete social

skill steps were demonstrated by the participant. For all steps to be recorded as correct, all elements of a step must have been demonstrated, and steps must have been demonstrated in the correct sequence. If a participant did not initiate a response within 10 s, a graduate researcher recorded all steps as inaccurate. A minimum of 30 s was required between cues of social skills.

Interobserver Agreement

An independent observer coded a minimum of 30% of opportunities to demonstrate target skills following a cue within each phase for all participants and skills. Interobserver agreement (IOA) was calculated by dividing the total number of agreements for skill steps demonstrated by the total number of agreements plus disagreements and multiplied by 100. Prior to the study, a graduate researcher was trained to code skill accuracy by the primary researcher by reviewing skill steps and recording accurate demonstrations of steps during role plays. The graduate researcher was required to demonstrate 100% IOA during training to serve as the independent observer. If agreement between the observers fell below 80% at any point, observers were retrained. Agreement only fell below 80% on one occasion ($M = 99.81\%$, range = 75%–100%). Following the instance during which agreement fell below 80%, the observer was retrained in operational definitions of skill steps.

Treatment Integrity

A checklist (see Table 4) was utilized during all phases to ensure the intervention was being implemented with fidelity. The checklist was adapted from the Superheroes Social Skills manual and described all elements of the intervention. The checklist was completed by the facilitator and an independent observer during each session, with intervention elements being marked as completed, not completed, or not applicable. Overall, integrity was reported to be

Table 3. Cues and Task Analyses of Target Social Skills.

Target skill	Conversation	Emotions	Responding to questions
Cue	“Hey (name of participant)”	Make sad/happy/mad face near participant	“What is your favorite (blank)?”
Step 1	Uses FEVER (Face the Person, Eye Contact, Appropriate Voice, Expression, and Relaxed Posture)	Face the person	Uses FEVER (Face, Eye Contact, Appropriate Voice, Expression, and Relaxed Posture)
Operational definition	<i>Head and shoulders oriented toward person, eye contact established for 3 s, voice at conversational level, face and body relaxed</i>	<i>Head and shoulders oriented toward person</i>	<i>Head and shoulders oriented toward person, eye contact established for 3 s, voice at conversational level, face and body relaxed</i>
Step 2	Greet the person and say something about a topic	Make eye contact	Listen to the person, as indicated by maintaining orientation and eye contact while the person is talking
Operational definition	<i>Relevant greeting provided, provides a comment that encourages engagement from person</i>	<i>Establish eye contact for a minimum of 3 s</i>	<i>Eye contact not broken for longer than 3-s intervals, head and body oriented toward person</i>
Step 3	Maintain eye contact while the other person speaks	Verbally label the emotion for yourself	Provide a contextually appropriate response
Operational definition	<i>Eye contact not broken for longer than 3-s intervals</i>	<i>States an emotion verbally (e.g., You feel sad, You are happy, You look mad)</i>	<i>Provides a response that is appropriate and related to the question</i>
Step 4	Make a comment about what the person said or ask a relevant question	N/A	Maintain eye contact while responding
Operational definition	<i>Provides a contextually relevant comment or question</i>	N/A	<i>Eye contact not broken for longer than 3-s intervals</i>

Note. Examples and non-examples of skills available from the authors upon request.

Table 4. Treatment Integrity Checklist.

Step	Video-only	Full curriculum
1	Collect baseline data	Collect baseline data
2	Collect maintenance data	Collect maintenance data
3	Play group rules video	Review group rules
4	Play Fast Hands video	Introduce target skill, skill steps, and provide rationale
5	Play peer model video	Pass out self-monitoring cards
6	Play Digital Comic	Play Fast Hands video
7	Collect target skill data	Play peer model video
8		Model accurate and inaccurate skill use
9		Role-play skill with participants
10		Provide feedback regarding skill use
11		Play Digital Comic
12		Play social game
13		Collect target skill data
14		Conduct Scooter Card drawing for reward
Total	___/7	___/14

100% across all phases of the study. IOA data indicated 100% agreement between raters.

Design and Data Analysis

A multiple probe design across behaviors with concurrent replication across participants was utilized to evaluate the

effects of the intervention on skill accuracy. Design elements adhered to current single-case design standards with respect to frequency and timing of data collection (What Works Clearinghouse, 2017). Specifically, the study incorporates a minimum of six phases with at least five data points per phase, has initial preintervention data collection sessions that overlap vertically, probe points immediately

prior to introducing the independent variable, and probe points for skills not receiving intervention simultaneously to another skill receiving intervention or meeting the intervention criterion. Skill accuracy in this study was assessed across three phases: baseline, intervention, and maintenance. Phase changes were contingent upon the majority of participants in the group demonstrating stable responding across three consecutive opportunities to demonstrate target skills. Delivery of cues for skills continued to be given following termination of instruction of a particular target skill to evaluate maintenance of the skill.

Treatment decisions and intervention effects were primarily analyzed via visual analysis. Specifically, visual analysis assessed changes in trend, level, and variability across phases, as well as immediacy of effect, overlap across phases, and consistency of effect.

Baseline. Target social skills were chosen based on the ASSP, which was completed by parents and caregivers prior to the start of the intervention. ASSP results were analyzed to find common deficits in skills across all participants and were used to inform what target skills were selected from the Superhero Social skills program. For example, “Maintains Eye Contact During Conversations” on the ASSP was scored low across all participants. As such, the Body Basics skill from the Superheroes Social Skills program was selected as a target skill. Based on ASSP data, skills from the Superheroes Social Skills manual that were selected included Body Basics, Conversation/Topic Maintenance, Recognizing Emotions in Yourself and Others, Participate, and Responding to Questions and Requests. During baseline, participants had free access to play materials and to engage with one another. Probes of skill accuracy were administered individually with a researcher to ensure consistency in probe delivery. A minimum of five cues per skill was presented by researchers without the provision of error correction or praise. Skill accuracy was recorded based on the participants’ response to the researcher’s cue. Participants were thanked for compliance following the presentation of all cues.

Video only. Intervention sessions during the Video-Only phase were approximately 20-min long and incorporated only the video-based components of the Superheroes Social Skills manual. Sessions occurred in a group format, with all participants receiving simultaneous instruction. Researchers began each session by welcoming the participants and playing the group rules video. Next, the researchers stated the name of the new target skill and played the videos associated with the identified skill. The first video consisted of animated superhero characters introducing the target skill and providing a rationale for skill use and was approximately 5 min in duration. The second video included three video models in which unknown peers demonstrated the

target skill accurately in various settings and was approximately 5 min in duration. The last video was an animated comic book that depicted the superhero characters using the target social skill and was approximately 5 min in duration. Although the Superheroes Social Skills program includes hard copies of comic books to distribute to participants, they were not provided to participants during this phase. Following the videos, participants were allowed free time with access to play materials. After 5 min of free play, researchers provided cues for target skills. Cues were identical to baseline, as participants were thanked for their participation but were not provided with error correction or praise. A minimum of five data points was collected during this phase. Participants were then dismissed from the group to their parent or guardian.

Full curriculum. Following data collection for the Video-Only phase, participants began the Full Curriculum phase. As with the previous condition, all sessions occurred in a group format. Intervention sessions during the Full Curriculum phase were approximately 45-min long and incorporated all components of the Superheroes Social Skills program. During each intervention session, researchers began by welcoming the group to the session and reviewing the group rules and behavior management system. The group rules video was not played during this phase, but rather explained verbally by the graduate researchers.

Following the review of rules and the behavior management system, participants were introduced to the target skill. Graduate researchers then provided the rationale for the skill use from the intervention manual and the discrete skill steps per the task analysis. Self-monitoring cards were then distributed to the group. Next, participants viewed the exact same introduction and peer models videos used in the previous phase. The researchers then demonstrated live models of the target skill, first with an inaccurate model in which the participants identified the inaccurate discrete steps followed by an accurate model in which the participants identified correct steps. Participants then engaged in role play of the target skill, where they had the opportunity to demonstrate the target skill with a researcher and receive feedback. Specifically, any instance of inaccurate skill use resulted in the provision of corrective feedback that corresponded to the inaccurate skill steps. Any instance of accurate skill demonstration (i.e., 100% skill acquisition) resulted in the participant filling in a bubble on their self-monitoring card. Each participant was required to demonstrate the target skill with 100% accuracy a minimum of three instances before role play was completed. Following role play, the researchers distributed a hard copy of the comic book and played the corresponding animated comic book. Afterward, participants played a game that required the use of the target skill (e.g., Simon Says for the Following Directions skill). Participants were then provided 5 min of

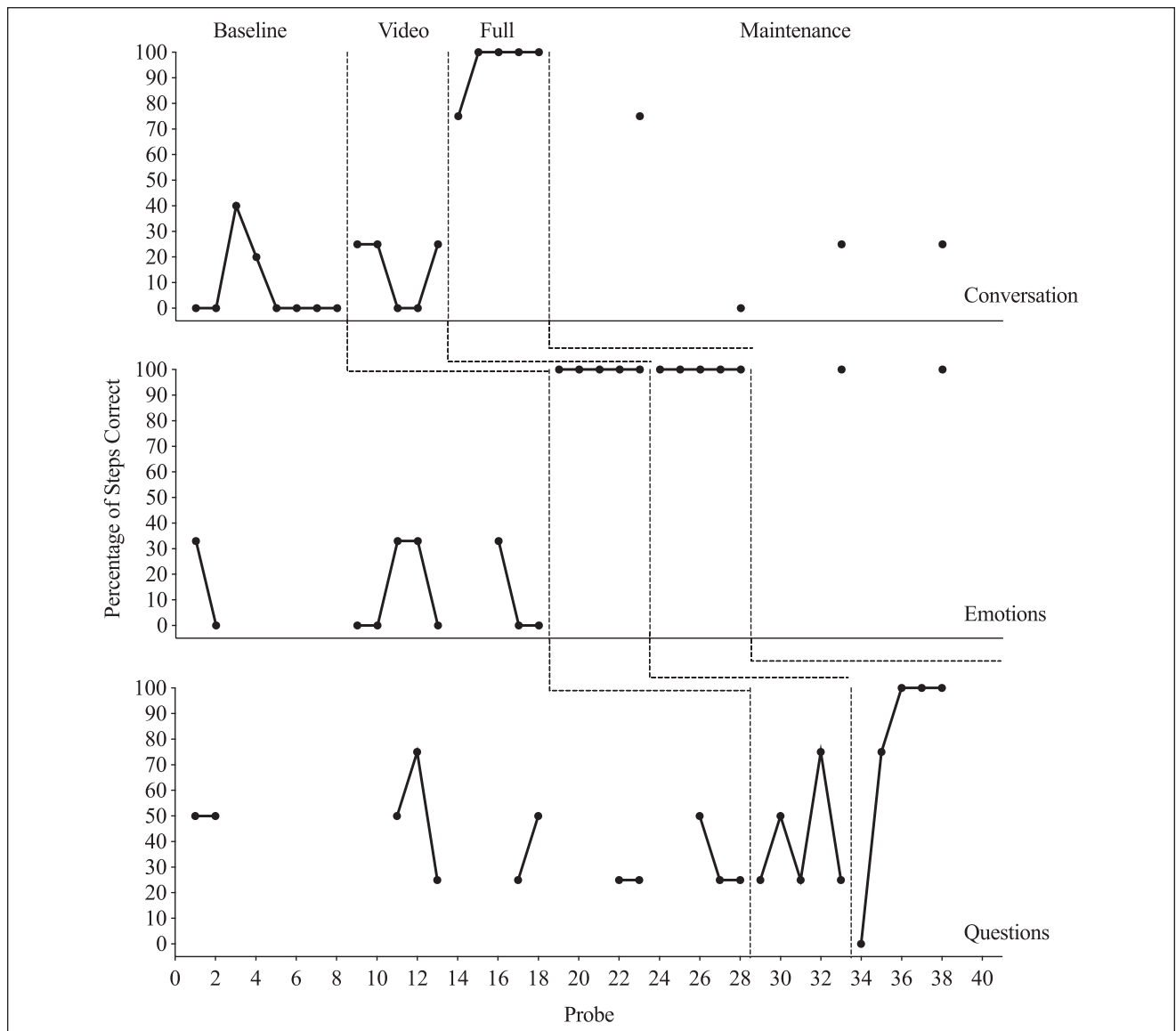


Figure 1. Percentage of steps correct, Otto.

free time, followed by data collection which looked identical to previous phases. After data collection was completed, a Scooter or Black Hole card was drawn and participants were provided with the corresponding consequence, self-monitoring cards were collected, and participants were dismissed from the group.

Maintenance. Data collection continued after skills were trained in both the Video-Only and Full Curriculum phases. Maintenance data collection began during the session immediately following termination of training of a particular target social skill. A minimum of one cue for skill use was presented during each session during the maintenance phase. Cues were identical to baseline in that they were not

followed by provision of error correction or praise with respect to skill accuracy.

Results

Otto

During baseline phases, Otto demonstrated variability and low levels of skill accuracy for Conversation ($M = 7\%$, range = 0%–40%) and Emotion ($M = 13\%$, range = 0%–33%), and low-to-moderate levels of skill accuracy for Responding to Questions ($M = 39\%$, range = 25%–75%; see Figure 1). Following implementation of the Video-Only phase, Otto continued to demonstrate low levels of

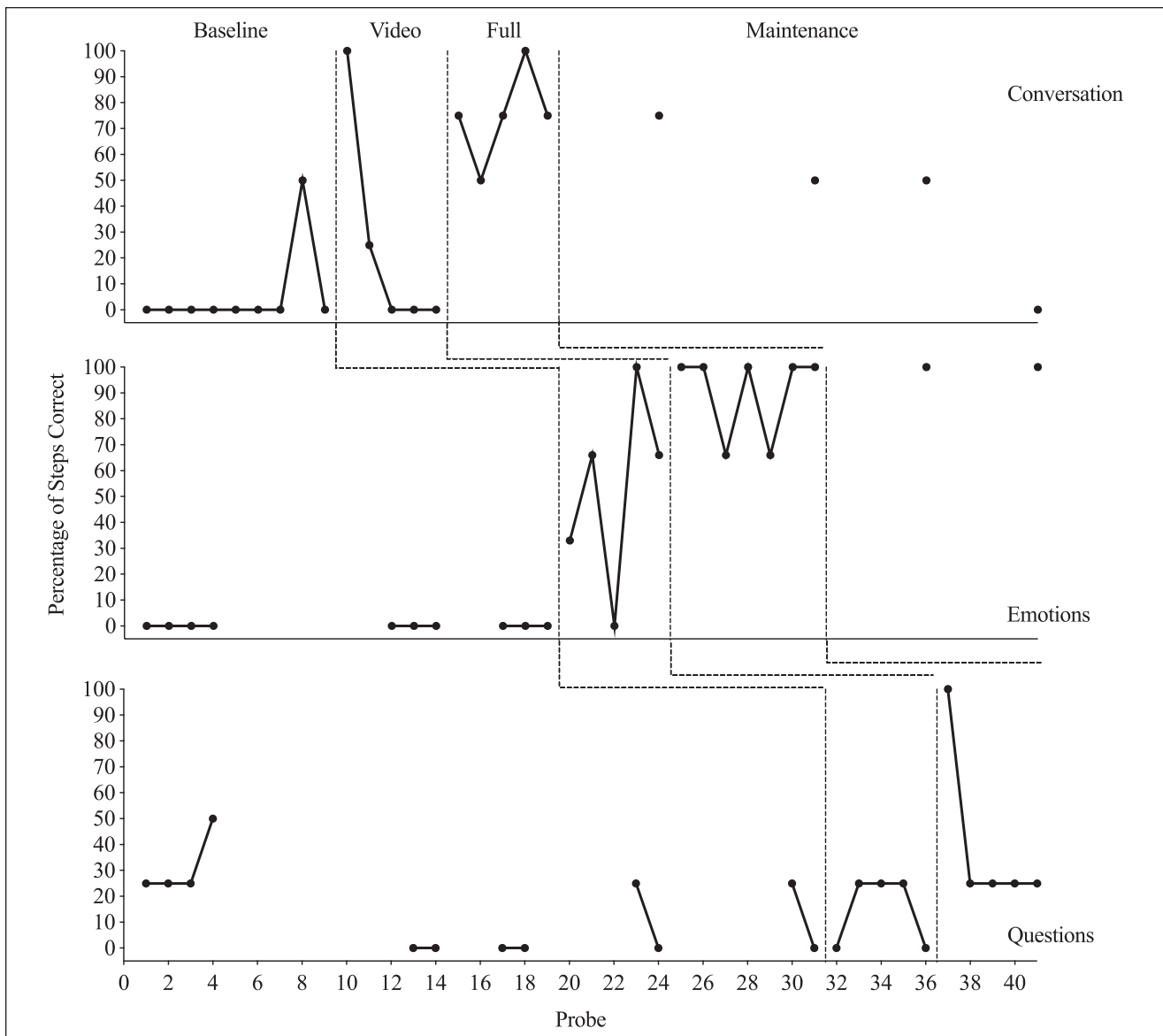


Figure 2. Percentage of steps correct, Lars.

skill accuracy for Conversation ($M = 15\%$, range = $0\%–25\%$). An immediate increase in skill accuracy was observed for Emotions ($M = 100\%$), with similar level of accuracy for Questions ($M = 55\%$, range = $25\%–100\%$). These results reveal an overall increase in mean skill accuracy across each target skill when compared to baseline performance. Following the Full Curriculum phase, Otto demonstrated an immediate increase in skill accuracy for Conversation ($M = 95\%$, range = $75\%–100\%$), with continued high levels of skill accuracy for Emotions ($M = 100\%$). Questions ($M = 75\%$, range = $0\%–100\%$) demonstrated increasing skill accuracy before stabilizing at high levels. During the maintenance phase, Otto demonstrated a decrease in skill accuracy for Conversation ($M = 31\%$,

range = $0\%–75\%$) with continued high levels of skill accuracy for Emotions ($M = 100\%$).

Lars

During baseline, Lars demonstrated low levels of skill accuracy across all skills, Conversation ($M = 5\%$, range = $0\%–25\%$) Emotions ($M = 0\%$), and Responding to Questions ($M = 14\%$, range = $0\%–50\%$; see Figure 2). During the Video Phase, Lars demonstrated an immediate increase followed by a sharp decrease in skill accuracy for Conversation ($M = 25\%$, range = $0\%–100\%$). Moderate-to-high levels of skill accuracy were observed for Emotions ($M = 59\%$, range = $0\%–109\%$), with continued low levels of skill

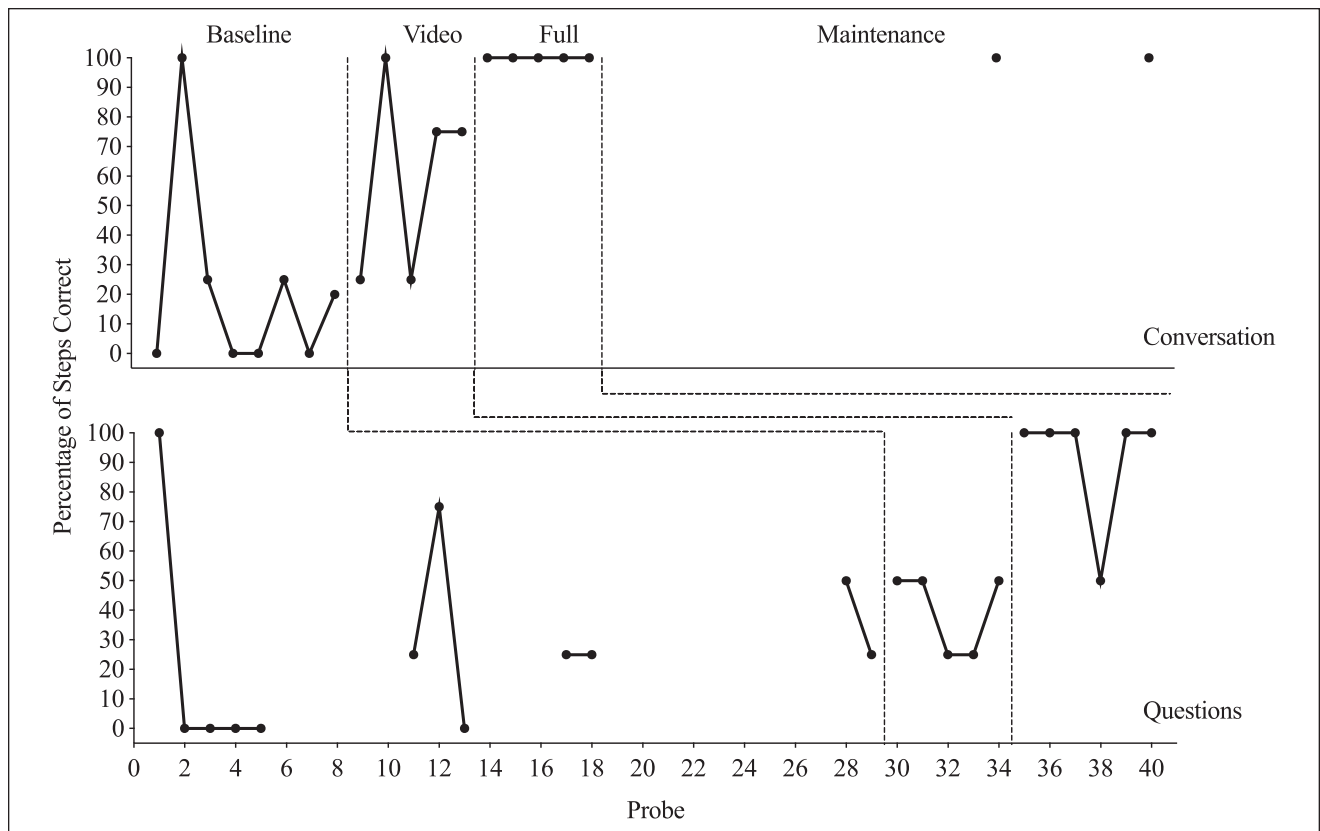


Figure 3. Percentage of steps correct, Trish.

accuracy for Questions ($M = 15\%$, range = 0% – 25%) following implementation of Video Only. Following the Full Curriculum phase, Lars demonstrated an immediate increase in skill accuracy for Conversation ($M = 75\%$, range = 50% – 100%), with the overall level increasing for Emotions ($M = 90\%$, range = 66% – 100%). Although skill accuracy immediately increased for Questions, improvements were not maintained throughout the phase ($M = 40\%$, range = 25% – 100%). During maintenance, Lars demonstrated high levels of skill accuracy for Emotions ($M = 100\%$) and moderate-to-low levels of skill accuracy for Conversation ($M = 43\%$, range = 0% – 75%).

Trish

During baseline, Trish demonstrated low-to-moderate and variable levels of skill accuracy for Conversation ($M = 21\%$, range = 0% – 100%) and Responding to Questions ($M = 27\%$, range = 0% – 100% ; see Figure 3). Following introduction of the Video-Only phase, an increase in level for skill accuracy was observed for Conversation ($M = 65\%$, range = 25% – 100%), with data being highly variable. Skill accuracy remained unchanged for Questions ($M = 40\%$, range = 25% – 50%). During the Full Curriculum

phase, variability in Conversation was eliminated and accuracy was high throughout the phase ($M = 100\%$). Immediate increases in skill accuracy were observed for Questions ($M = 91\%$, range = 50% – 100%). During the maintenance phase, Trish maintained high levels of skill accuracy for Conversation ($M = 100\%$).

Tito

Tito demonstrated low levels of skill accuracy for Conversation ($M = 2\%$, range = 0% – 25%), low-to-moderate levels of skill accuracy for Emotion ($M = 33\%$, range = 0% – 66%), and moderate levels of skill accuracy for Responding to Questions ($M = 51\%$, range = 0% – 100% ; see Figure 4). Following implementation of the Video-Only phase, Tito demonstrated low levels of skill accuracy for Conversation ($M = 25\%$), and moderate and unchanged levels of skill accuracy for Emotions ($M = 59\%$, range = 33% – 66%) and Responding to Questions ($M = 60\%$, range = 25% – 75%). Following implementation of the Full Curriculum phase, Tito demonstrated immediate and stable increases in skill accuracy across all skills: Emotions ($M = 100\%$), Conversation ($M = 100\%$), and Questions ($M = 95\%$, range = 75% – 100%). During maintenance,

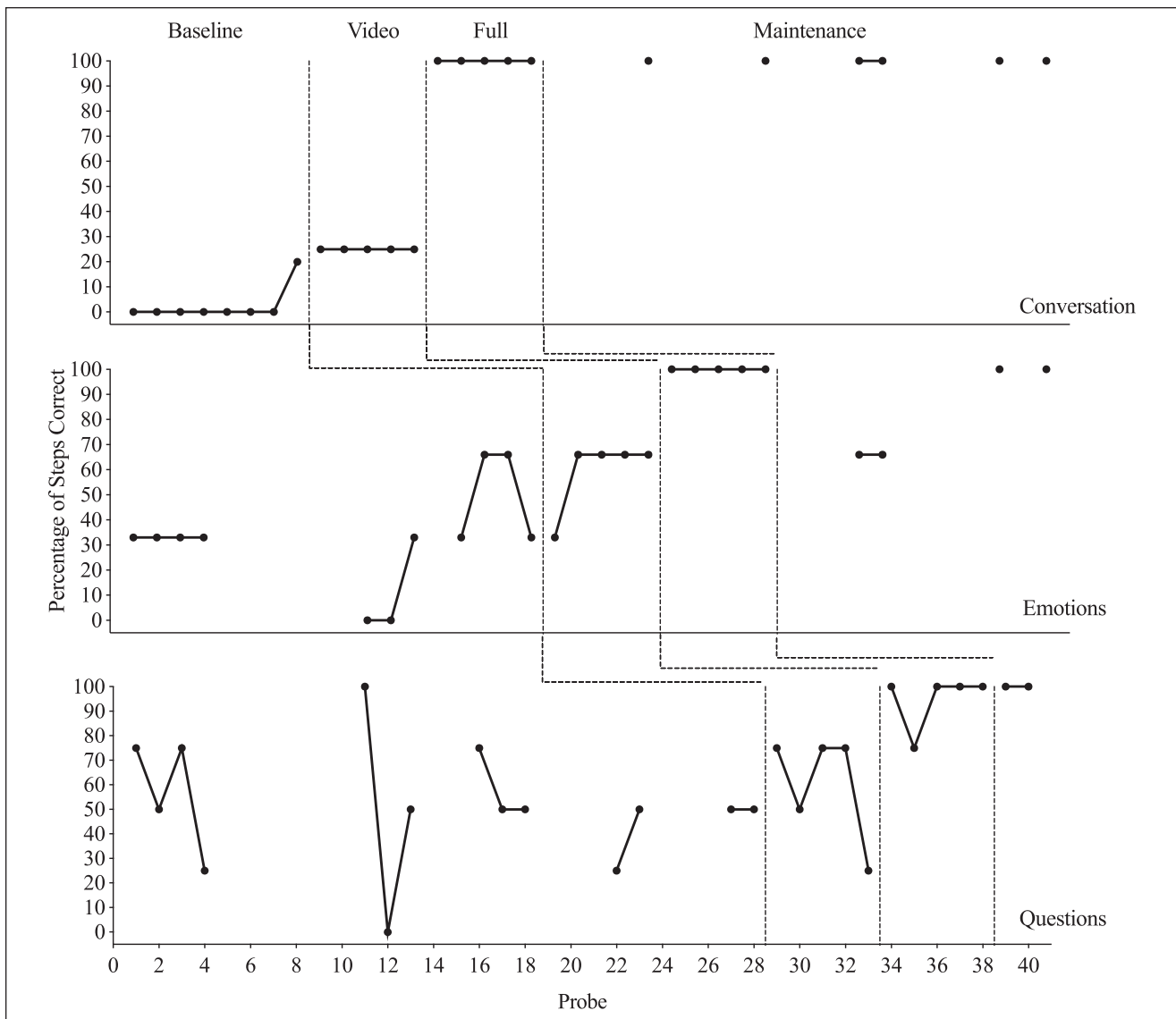


Figure 4. Percentage of steps correct, Tito.

Tito maintained high levels of accuracy for Conversation ($M = 100\%$), with an initial drop in accuracy for Emotions ($M = 83\%$, range = 66%–100%).

Discussion

The purpose of this study was to compare the efficacy of video-based elements of Superheroes Social Skills in isolation to the implementation of the full curriculum. Across all participants, implementation of the video components alone was associated with improvements in skill accuracy. However, in all but one case (i.e., Otto, Emotions),

implementation of the full curriculum was associated with further improvements in skill accuracy.

Previous reviews and meta-analyses have found video modeling to meet criteria as an evidence-based practice (e.g., Qi et al., 2018; Wang et al., 2011). In considering the lack of substantial improvements in the percentage of steps demonstrated under video only conditions, it is critical to consider differences in this study and previous research. First, studies included in previous reviews and meta-analyses, as well as current research, have focused on social outcomes such as duration of social engagement (e.g., Nikopoulous & Keenan, 2007), number of scripted

behaviors (e.g., Ezzeddine et al., 2020; MacDonald et al., 2005), and frequency of imitations (e.g., Cardon & Wilcox, 2011). However, limited research has evaluated whether video modeling may result in accurate demonstration of a multistep social skill as in this study. Despite researchers having found video modeling to produce improvements in demonstration of multistep vocational skills (e.g., Bidwell & Rehfeldt, 2004; Wu et al., 2016), differences in social outcomes examined in this study and previous research may account for some differences in results. Another possible explanation for the differences observed may be the dosage of video modeling in this study and in previous research. In this study, video-only intervention sessions were 20 min in duration; however, only 5 min were devoted to traditional video modeling in which same-age peers demonstrated target skills. The remainder of the time consisted of animated superheroes presenting skill rationale and steps to the skill, as well as an animated social narrative in which the target skill was utilized. Thus, the relative amount of time dedicated to video modeling may have been insufficient to produce substantial improvements in the percentage of skill steps demonstrated.

Although the results of the study indicated that video elements alone were often insufficient to promote behavior change, participants consistently demonstrated high levels of skill accuracy when the full Superheroes Social Skills curriculum was implemented. These findings are consistent with previous research finding participation in the curriculum to result in improved skill accuracy of participants with ASD (e.g., Murphy et al., 2018; Radley et al., 2014, 2016). Together with previous findings, this study adds to a body of literature suggesting that the intervention is effective in promoting the use of target skills as task analyzed.

The findings of this study have important implications for practitioners. Implementation of the video elements alone represents a less time- and resource-intensive intervention that is capable of promoting behavior change. However, these changes were often small and readily improved upon when the full curriculum was implemented. As such, these data suggest that practitioners should not consider the video elements in isolation to be sufficiently effective to stand as an intervention by themselves. This does not, however, suggest that the video elements of the intervention are unnecessary. Specifically, the Full Curriculum phase of the study included both video-based and non-video-based elements, with non-video-based elements never being presented in isolation. As such, the sufficiency of the non-video-based elements in isolation remains unknown—however, their necessity is apparent given that consistently high levels of accuracy were only observed in the Full Curriculum phase. As such, practitioners should consider implementation of the full curriculum to ensure maximal benefit.

Regarding implications for researchers, this study highlights the need for further component analyses of the Superheroes Social Skills program. Within the Superheroes Social Skills program, many individual elements comprise the video and non-video elements of the intervention. For example, individual elements included only as part of the Full Curriculum phase of this study consist of behavioral skills training and a social game. Although previous research has highlighted the effectiveness of behavioral skills training in promoting social skills (Stewart et al., 2007), it is unknown if the social game provided any additive benefit or if similar effects would be observed if removed from the intervention.

Although many multicomponent social skills interventions for individuals with ASD have been developed, few have undergone an analysis to determine active elements of the intervention. As such, it is possible that practitioners are needlessly implementing elements of intervention that confer little additive benefit (Barnett et al., 2004; Deitz et al., 1983). Relatedly, as practitioners may deviate from an intervention as described by a treatment manual, it is critical to identify those elements that are essential to behavior change. The completion of analyses such as the current study is essential to improving the efficiency with which social skills interventions are implemented and is likely to increase the social validity of these interventions (e.g., Ward-Horner & Sturmey, 2012).

Limitations

This study is not without limitations. First, the Superheroes Social Skills program comprises many intervention elements. In this study, elements were categorized as either video-based or not video-based. However, video-based components comprise instruction by animated superheroes, video models of unknown peers, and an animated social narrative. Non-video-based components include behavioral skills training, use of a self-monitoring card, and a social game. As such, the individual contribution of each of the elements within the video-based or not video-based categorizations is unknown. As individual components were not implemented in isolation, this study is only able to identify elements that are necessary but cannot describe the sufficiency of any one element to produce marked improvements in skill accuracy. Future researchers may consider conducting a more detailed additive analysis of intervention elements as a means of determining the relative contribution and necessity of each strategy included within the Superheroes Social Skills program. Alternatively, elements may be implemented in isolation to determine whether use alone would be sufficient for promoting behavioral change.

Relatedly, this study is limited by order effects. Specifically, the phases were implemented in the same order

across all participants, with each experiencing the video-based elements in isolation prior to receiving the full curriculum. Thus, it is unknown whether the high levels of skill accuracy observed during the Full Curriculum phase were a result of the non-video-based elements or the combination of all elements. Put another way, it is unknown whether high levels of skill accuracy would have been immediately observed had the non-video-based elements been implemented as the first intervention condition. To better identify the relative contribution of non-video-based elements, future researchers should consider counterbalancing phases.

Other limitations that should be considered include one participant (Trish) being absent from intervention for the Emotions skill due to vacation. Related to participant data, no standardized assessment data were collected for Tito due to lack of consent. Tito demonstrated social difficulties that were similar to those of the other participants, but due to the lack of assessment data, it is unknown regarding the degree to which Tito represented a typically developing peer.

We did not assess all of the relevant variables that could have provided more insight into our results. Specifically, we did not assess the social validity of the Superheroes Social Skills program. It may be possible that participants' data were partially explained by their like or dislike of the procedures. Similarly, we did not assess the generalization of treatment effects across setting or person, limiting the extent to which the results can be expected to maintain in the absence of training stimuli. Previous research on the Superheroes Social Skills program has supported its social validity and ability to produce generalizable effects (e.g., Radley et al., 2014); however, those findings may not have held true in this study.

Conclusion

Although previous research has found Superheroes Social Skills program to be useful in promoting acquisition and use of target social skills, the multicomponent nature of the intervention limited the ability of researchers to identify elements that are necessary for behavior change. The results of this study demonstrate that video-based elements of the program are insufficient to consistently promote high levels of skill accuracy. Although implementation of the video-based elements in isolation represents a less resource-intensive intervention, their limited efficacy in promoting behavior change is a substantial drawback. Although additional research is needed to further identify the relative contribution and sufficiency of non-video-based elements of the intervention, the data collected within this study suggest that participants receiving the Superheroes Social Skills intervention may obtain maximal benefit when exposed to the full curriculum.

Declaration of Conflicting Interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Funding

The author(s) received no financial support for the research, authorship, and/or publication of this article.

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